

PROFESSIONAL SUMMARY

Software Engineer with 2 years of experience building scalable backend systems and Agentic AI workflows at Qualcomm. Experience in developing ML orchestrators, production RAG pipelines, and cloud-native applications using Python, FastAPI, Kubernetes, and modern LLM frameworks. Strong foundation in data structures, system design, and performance optimization.

EXPERIENCE

- Qualcomm

July 2024 – Present

Software Engineer / Python, FastAPI, GenAI, RAG

- Designed distributed ML orchestration service using Python and FastAPI, enabling asynchronous execution of GenAI workflows and improving processing throughput by 40%.
 - Optimized production RAG pipelines by improving retrieval logic, vector search performance, and database architecture, reducing end-to-end inference latency by 15%.
 - Automated QA workflows by scripting a robust data translation module mapping Orbit customer issues to automated test cases, significantly accelerating the CI/CD pipeline.
- Oracle | Python, OCI, OADB, SQL

May 2023 – July 2023

Summer Intern

- Architected a Python utility leveraging Oracle Object Storage to securely ingest, process, and store 300 GB of unstructured data daily.
 - Integrated object storage systems with Oracle Autonomous Database, streamlining data access and improving SQL query performance on large-scale datasets.
 - Spearheaded the development of real-time tracking dashboards in Oracle APEX, reducing manual team reporting requests by 20%.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: PyTorch, Hugging Face Transformers, FastAPI, ReactJS
Backend AI Engineering: Distributed Systems, REST APIs, ML Orchestration, RAG Pipelines, LLM Architectures, Vector Databases
Cloud & Databases: OCI, Docker, Kubernetes, MongoDB, Oracle Autonomous DB
Developer Tools: Git, VS Code, Oracle APEX

EDUCATION

- National Institute of Technology, Warangal

2020 – 2024

B.Tech, Electronics and Communication Engineering (CGPA: 7.48)

SELECTED PROJECTS

- Enterprise Document RAG Assistant

LangChain, OpenAI GPT-4o, Pinecone, RAGAS

- Architected a Retrieval-Augmented Generation (RAG) knowledge assistant over 500+ technical PDFs using LangChain, OpenAI GPT-4o, and a Pinecone vector database.
 - Adopted a two-stage retrieve-and-rerank architecture with semantic chunking to manage context windows, improving retrieval recall by 30% on complex queries.
 - Developed a comprehensive evaluation harness using the RAGAS framework to systematically identify and log retrieval failures versus generation failures, achieving an 88% context precision score.
- Multi-Agent Research Workflow

LangGraph, Anthropic Claude 3.5

- Built an autonomous multi-agent research system using LangGraph and Anthropic Claude 3.5 to orchestrate specialized planner, researcher, and writer agents.
 - Implemented stateful agent memory and conditional routing logic, reducing manual research synthesis time by 40% while enforcing strict token cost ceilings per run.
 - Integrated a human-in-the-loop review step for external API tool executions, catching hallucinated citations and preventing material errors in 15% of draft outputs before final generation.

CERTIFICATIONS & ACHIEVEMENTS

- CodeChef 4-Star (Max Rating: 1824)

•LeetCodeSolved 500+ algorithmic and data structure problems

•WSDC HackathonSecured 6th place in Web Development category